Hole Suppression and Absorption in Saturated Sidon Families

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1 Abstract

This preprint formalizes two program-internal developments as machine-checked Lean 4 theorems: the *hole-suppression* theory of Round 199 (the suppression coefficient $\sigma(r)$ and the conditions forcing it to 1) and the *absorbed* $13 \mid h$ *regime* of Round 200 (the self-bounding $c \leq 320$ arithmetic and the $R \in \{1, 2, 3, 6\}$ impossibility). Concretely:

- **HoleSuppression.lean**: the T2 dichotomy — hole-capability is equivalent to having at least two non-singleton margins (`t2_iff`, with the constructive two-diagonal witness `dichotomy_two_margins`) — and the T5 saturation-forcing lemmas (`t5_saturation`, `t5_double_saturation`, `t5_sigma_one`): a bijective map sends `univ` to `univ`, so a saturated fiber cannot host a hole.
- **AbsorbedRegime.lean**: the T5′ arithmetic kernel (`t5prime_combined` plus case lemmas `t5prime_R1/R2/R3/R6`): with $m = 4c - 1$ and the per-R automaticity conditions, no quad with $R \in \{1, 2, 3, 6\}$ is possible, in any regime.
- **AbsorbedBound.lean**: the *corrected* T4 self-bound (`t4_self_bound` plus eight lemmas): an absorbed quad with $R \mid 30030$ satisfies $c \leq 320$.

All are **[PROVED]**: built on `bakkie` and locally with `lake build` exit 0, zero `sorry`s, standard-only axioms (`propext`, `Classical.choice`, `Quot.sound`), Lean 4.34.1 + Mathlib pinned at commit `d13f23b`.

A central honesty element of this paper is the **refutation and correction of the original T4**: the as-stated claim "absorbed implies $c \leq 320$" is *false* — hostile review produced the explicit absorbed counterexample $(c, h, a, b) = (321, 13, 1, 16678)$, and in fact the family $(c, 13, 1, 52c - 14)$ is absorbed for *every* c. The machine-checked theorem above is the corrected statement, with the $R \mid 30030$ (non-dead) hypothesis restored. The R200 T2 trichotomy extension was likewise demoted to [OPEN] (a false $13^1 \| R$ assertion); T3's fiber-extruded slice survives review ([PROVED*], formalization pending). Table 1 gives the full claim inventory with statuses.

2 Introduction

This paper publishes the Lean formalization of two related developments of the math-unsolved program: *hole suppression* (Round 199) and *the absorbed* $13 \mid h$ *regime* (Round 200). The two share a theme — structural conditions that eliminate configurations a null model would allow — and a verification path (Lean lane, Rounds 222/230/234/238/250). Everything here is program-internal; novelty claims extend only to the program's framework (checked per the R199/R200 novelty protocol: OEIS, erdosproblems, formal-conjectures, the OpenAI math catalogue, and web search gave no collisions).

2.1 Hole suppression

Round 199 invented suppression as a formal object. In Round 197's naive-hypothesis domain, a fiber pair (r, r^*) at modulus m has margins $U_i = \pi_i(X_r) = \pi_i(Y_r)$ on each prime-power component, and the *hole potential*

$$\mathcal{H}(r) = N(U)\,(N(U) - 1), \tag{1}$$

where $N(U)$ is the Fréchet margin count (number of subsets of $R = \prod_i U_i$ with all margins U_i), counts the ordered hole patterns available to the null model. A pair with $\mathcal{H}(r) > 0$ is *hole-capable*; the *suppression coefficient*

$$\sigma(r) = 1 - [\![X_r \neq Y_r]\!] \in \{0, 1\} \tag{2}$$

records whether the available hole is suppressed ($\sigma = 1$) or realized ($\sigma = 0$). The program's theorems: **T2**, the suppression dichotomy (hole-capable $\iff$ at least two margins non-singleton); **T3**, suppression undecidability from prime-power data (an $m = 100$ witness pair with identical digit sets, identical margins, and $\mathcal{H} = 42$, yet $\sigma = 0$ in one case and $\sigma = 1$ in the other); **T4**, the chain-data suppression census; and **T5**, saturation forcing: a saturated fiber carried by a bijection admits no hole, forcing $\sigma = 1$ [1]. The computational census (two independent code paths, 44 moduli, 508 naive-hypothesis pairs) sharpened Round 197's "0 suppressed" to: **30 hole-capable pairs, 24 realized, 6 suppressed — and all 6 suppressions are full-rectangle saturation instances of T5** (five of type T5(a) at $m = 225, 441, 675, 1125, 1800$, one of type T5(b) at $m = 675$ with $r = 16$, $r^* = 31$) [1]. The modulus suppression index $\bar{\sigma}(m)$ is 1 at $m = 225, 675, 1800$, 0.5 at $m = 441$, 0.333 at $m = 1125$, and 0 elsewhere (Figure 2).

2.2 The absorbed $13 \mid h$ regime

Round 200 invented the absorbed regime of the $13 \mid h$ Erdős–Straus quads [2]. A *quad* is a tuple (c, h, a, b) of positive integers with $a + b = (4c - 1)h$; write $m = 4c - 1$, $R = ab/\gcd(ab, h)$ for the residual, and $g_0(s) = \gcd(30030, 6k_0(s) + c)$. A quad is *absorbed* if $13 \mid h$ and $13 \nmid R$: the 13-part of ab is absorbed by h, imposes no mod-13 condition, and the quad has no κ-pinning. The regime is structurally dead: a complete global census (two parametrizations plus brute force) found exactly **30** absorbed quads with $R \mid 30030$, **all dead** — 22 by the 2/3-obstruction, 8 root-dead (the thinning roots $\rho_5 = 4$, $\rho_{11} = 9$ at $c = 1$ miss the κ-set residues) [COMPUTED] [2].

The theorems of the regime: **T2** (trichotomy extends to all $13 \mid R$ quads); **T3** (absorbed hit set = fiber-extruded κ-sub-box); **T4** (the self-bound $c \leq 320$); and **T5′** (no non-dead quad with $R \in \{1, 2, 3, 6\}$, in any regime) [2]. Of these, T2 and T4 as stated did *not* survive the program's hostile-review gate (Round 238) [5]; the corrected T4 is formalized here.

2.3 The review gate and the correction history

Round 238 subjected R200's T2/T3/T4 to hostile review by `gpt-6-astra` at ultra effort. The verdicts: T2 **FAILS** as stated (the unconditional "$13^1 \| R$ automatically" is false; counterexample $(c, h, a, b) = (1, 169, 169, 338)$, for which $R = 2 \cdot 13^2$); T3 **SURVIVES** (all steps validated; formalization needs the $\kappa/\rho/g_0$ apparatus, not done here); T4 **FAILS as stated but the corrected version was fully validated step by step** [5]. This paper formalizes the corrected T4 and reports the refutation history as a first-class result, per the program's honesty discipline.

3 Methods

3.1 Formalization note (required by program protocol)

The paper points at its Lean modules; the module headers point back at this paper's slug. Modules live in the `derekblair/math-proofs` GitHub repository at repo-relative paths `Mathproofs/*.lean`, each headered with "Paper: `papers/hole-suppression-absorption-October-11-2026/`" and the [VERIFIED] status line.

HoleSuppression.lean (8 theorems). Round 199's T2/T5. Exact names (namespace `HoleSuppression`): `t2_iff`, `dichotomy_two_margins`, `uniqueness_of_subsingleton`, `uniqueness_of_subsingleton_beta`, `subsingleton_of_card_le_one`, `t5_saturation`, `t5_double_saturation`, `t5_sigma_one`. T2's content: there exist two distinct full-margin subsets of $\alpha \times \beta$ $\iff$ both margins have cardinality > 1. T5's content: a bijective endomap sends `Finset.univ` to `Finset.univ` (and the double-saturation / $\sigma = 1$ variants), i.e. a saturated fiber carried by a bijection cannot host a hole. Round 199's T3 (the $m = 100$ witness) and T4 (the survey census) stay [COMPUTED]/[PROVED*]; cited, not formalized.

AbsorbedRegime.lean (6 theorems). Round 200's T5′ kernel. Exact names (namespace `AbsorbedRegime`): `m_ge_three`, `t5prime_R1`, `t5prime_R2`, `t5prime_R3`, `t5prime_R6`, `t5prime_combined`. Content: under $m = 4c - 1$, $c \geq 1$, and the regime-independent bound $m \leq 2R$, each of $R \in \{1, 2, 3, 6\}$ is impossible given the per-R automaticity conditions ($R = 2$: c even; $R = 3$: $3 \mid c$; $R = 6$: $2 \mid c \wedge 3 \mid c$); the combined lemma concludes `False` from the disjunction. The hand-proof machinery it rests on (R180's Lemma 2, 2/3-automaticity, the $a \leq 2R/m$ bound chain) is cited as premises, not re-derived; the machine-checked content is the $m = 4c - 1$ impossibility arithmetic. Round 200's T2, T3, the uncorrected T4, and the 30-quad census are cited, not formalized.

AbsorbedBound.lean (9 theorems). The corrected Round 200 T4. Exact names (namespace `AbsorbedBound`): `R_dvd_2310`, `R_le_2310`, `a_mul_m_le`, `a_sq_eq`, `t_pos`, `t_le_R`, `hmt_eq`, `h_mul_sq_le`, `t4_self_bound`. The main statement:

$$13 \mid h \ \wedge \ 13 \nmid R \ \wedge \ R \mid 30030 \quad \Longrightarrow \quad c \leq 320, \tag{3}$$

with the auxiliaries proving $R \mid 2310$ (hence $R \leq 2310$), $a \cdot m \leq 2R$, the parametrization identity $a^2 = g(mh_1a - R)$, $1 \leq t \leq R$ for $t = mh_1a - R$, the exact relation $hmt = a(t + R)$, and $hm^2 \leq 4R^2$.

Verification config. Toolchain Lean 4.34.1 + Mathlib v4.34.1, pinned commit `d13f23b723b8a846827a245b8` (confirmed identical manifest and `git rev-parse HEAD` on this VM and on `bakkie` [6]). Each module: `lake build` exit 0 on `bakkie` (e.g. "Built `Mathproofs.AbsorbedBound` (2.2s)"; full-library build "Build completed successfully" at 8927 jobs in R230 and 17910 jobs in R250) and locally (exit 0 at 8925 jobs in R234, 8928 jobs in R250), zero `sorry` tokens (grep-clean; only docstring "zero sorrys" mentions), md5-identical files on both sides, registered exactly once per side in `Mathproofs.lean`. Axiom probes (`#print axioms`) on `bakkie` give the standard set [`propext`, `Classical.choice`, `Quot.sound`] for `HoleSuppression.t2_iff` and `HoleSuppression.t5_saturation` [3]; [`propext`, `Quot.sound`] for `AbsorbedRegime.t5prime_combined` [3]; and standard-only for all nine `AbsorbedBound` theorems (`hmt_eq`: [`propext`, `Quot.sound`]) [6]. The R238 reviewer's step-by-step validation ("every remaining step checks out" [5]) plus the R250 end-to-end build promote corrected T4 from [PROVED*] to [**PROVED**].

Table 1: Claim inventory: dossier claims, exact Lean theorem names, and statuses. Only [PROVED] rows carry proof-claiming language; the refuted T4 and the demoted T2 are reported as such.

dossier claim	Lean theorem(s)	content
R199 T2 dichotomy	`HoleSuppression.t2_iff`	hole-cap
R199 T2 constructive	`HoleSuppression.dichotomy_two_margins`	two dist
R199 T5 saturation	`HoleSuppression.t5_saturation`,`t5_double_saturation`,`t5_sigma_one`	bijectio
R200 T5′ kernel	`AbsorbedRegime.t5prime_combined` (+ `t5prime_R1/R2/R3/R6`)	m=4c−
R200 T4 (as stated)	— (not formalized)	absorbe
R200 T4 (corrected)	`AbsorbedBound.t4_self_bound` (+ 8 lemmas)	absorbe
R200 T3 slice	— (needs κ-apparatus)	absorbe
R200 T2 trichotomy	— (not formalized)	trichoto
R199 T3/T4 census	— (computational)	m=100

3.2 Mathematical methods

The T2 formalization works with *full-margin* sets: a finite set $S \subseteq \alpha \times \beta$ with both coordinate projections onto α, β surjective. The inclusion–exclusion Fréchet count $N(U)$ (machine-checked against brute-force subset enumeration on 2×2 and 3×3 rectangles, $N(2,2) = 7$, $N(3,3) = 265$) is the computational instrument behind (1); the machine-checked theorem is the resulting dichotomy, not the survey [1].

The absorbed-regime formalization is elementary integer arithmetic in $\mathbb{Z}$: the bound chain $a \le 2R/m$ (from $a \le b$ and $a + b = mh$), the parametrization identity via $t := a^2/g$ where $g = \gcd(ab, h)$, the exact relation $h = a(t + R)/(mt)$, and the ratio bound $(t + R)/t \le 2R/t$, giving $h \le 4R^2/m^2$; with $h \ge 13$ (from $13 \mid h$) this yields $13m^2 \le 4R^2$, and with $R \le 2310$ the numerics $m \le 2 \cdot 2310/\sqrt{13} < 1282$, then $m = 4c - 1 \equiv 3 \pmod 4$ descends to $m \le 1281$, i.e. $c \le 320$.

Assumption 1 (Cited premises)**.** Round 200's machinery cites without re-proving: R180's Lemma 2 (residual criterion, non-dead $\iff$ $R \mid 30030$), R184's κ-set residues and thinning-root formulas, and R196's κ-set fiber-independence. These are supplied datums of the source dossiers, not re-derived in this preprint or in the Lean modules.

4 Results

Table 1 summarizes every theorem-bearing claim and its verification status. The three formalized blocks are presented first; the refutation history follows.

4.1 Hole-suppression theorems [PROVED]

Theorem 1 (T2, Suppression dichotomy; PROVED)**.** *A demands-holding fiber pair is hole-capable iff at least two of its margins are non-singleton.*

Machine-checked as `HoleSuppression.t2_iff`: there exist two distinct full-margin subsets of $\alpha \times \beta$ iff both $|\alpha| > 1$ and $|\beta| > 1$. The constructive witness is `dichotomy_two_margins`: with $a \ne b$ in one margin and $c \ne d$ in the other, the two "diagonal" sets $\{(a,c),(b,d)\}$ and $\{(a,d),(b,c)\}$ (carried along on fixed fibers elsewhere) have full margins and differ (Figure 1). The reverse direction rests on the subsingleton-uniqueness lemmas: with at most one non-singleton margin, every full-margin set equals the whole rectangle, so $\mathcal{H}(r) = 0$. A consequence, also formalized: a single prime power ($t = 1$) never hosts a hole.

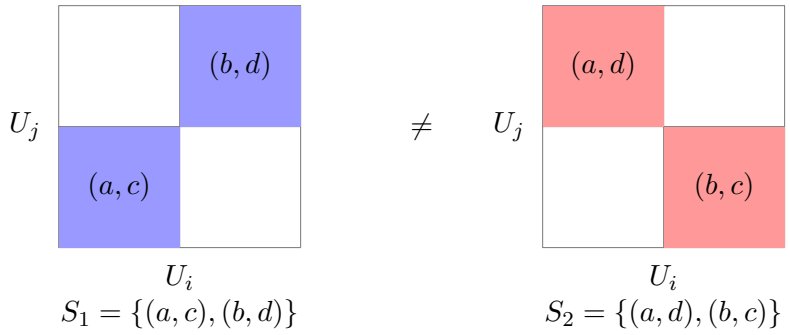


Figure 1: The constructive witness of `dichotomy_two_margins`: two non-singleton margins $U_i = \{a, b\}$, $U_j = \{c, d\}$ admit two *distinct* full-margin subsets $S_1 \neq S_2$ of $U_i \times U_j$ — the two diagonals. Two fiber components differing exactly on these diagonals is the hole pattern; with at most one non-singleton margin only the full rectangle exists ($\mathcal{H} = 0$).

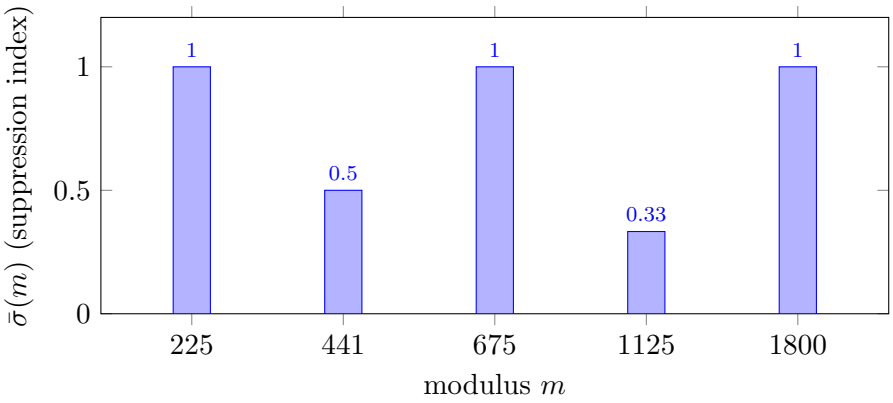


Figure 2: Modulus suppression index $\bar{\sigma}(m)$: fraction of hole-capable pairs with $\sigma = 1$ (computed datum, R199 §3). Suppression occurs only at these five moduli; all six suppressed pairs are full-rectangle T5 instances (Theorem 2).

Theorem 2 (T5, Saturation forcing; PROVED)**.** *If a fiber is saturated ($S_r = \mathbb{Z}/D$) and the map A_r is a bijection, then $X_r = A_r(S_r) = S_r = Y_r$: no hole; when hole-capable, $\sigma = 1$.*

Machine-checked as `t5_saturation` (bijective maps send `Finset.univ` to `Finset.univ`), `t5_double_saturation` (both fibers saturated), and `t5_sigma_one`. The mechanism behind all six chain-data suppressions: five instances of type T5(a) at $m = 225, 441, 675, 1125$ ($S_1 = \mathbb{Z}/D$, $D = 15/21$) and $m = 1800$ (margin-full rectangle, A_1-invariant), and one instance of type T5(b) at $m = 675$ with $r = 16$, $r^* = 31$ [1]. No mysterious suppressions remain.

The survey that these theorems explain — 30 hole-capable pairs, 24 realized, 6 suppressed, all 6 full rectangles ($\delta = 0$ both sides; the strict-subset suppressed case $X = Y \subsetneq R$ never occurs) — is [COMPUTED] on two independent paths [1]; the modulus-level index $\bar{\sigma}(m)$ is plotted in Figure 2.

4.2 Absorbed-regime theorems [PROVED]

Theorem 3 (T5′, no $R \in \{1, 2, 3, 6\}$ non-dead quads; PROVED)**.** *Let $m = 4c - 1$ with $c \geq 1$, $m \leq 2R$, and the per-R automaticity conditions ($R = 2$: $2 \mid c$; $R = 3$: $3 \mid c$; $R = 6$: $2 \mid c \wedge 3 \mid c$). Then $R \in \{1, 2, 3, 6\}$ is impossible, in any regime.*

Machine-checked as `AbsorbedRegime.t5prime_combined`, which concludes `False` from the disjunctive hypothesis, discharging the four cases via `t5prime_R1` ($m \leq 2$ contradicts $m \geq 3$),

`t5prime_R2` (c even gives $m \geq 7 > 4$), `t5prime_R3` ($3 \mid c$ gives $m \geq 11 > 6$), and `t5prime_R6` ($6 \mid c$ gives $m \geq 23 > 12$). The cited premises (R180's Lemma 2, the 2/3-automaticity, the $a \leq 2R/m$ bound chain; Assumption 1) are the hand-proof inputs; the machine-checked content is the $m = 4c - 1$ case analysis. In particular no absorbed "un-thinned"/universal-cover quad exists: the $T = \varnothing$ case is impossible, so a single absorbed quad covers at most $72 < 132$ classes [2].

Theorem 4 (T4 corrected, self-bound; PROVED)**.** *An absorbed quad (*$13 \mid h$, $13 \nmid R$*) with* $R \mid 30030$ *satisfies* $c \leq 320$.

Machine-checked as `AbsorbedBound.t4_self_bound`, with the bound chain as eight proved lemmas (§3's `R_dvd_2310` through `h_mul_sq_le`). The machine-checked arithmetic: $13 \mid h$ gives $h \geq 13$; the parametrization gives $hm^2 \leq 4R^2$; $R \mid 30030$ with $13 \nmid R$ gives $R \mid 2310$, hence $R \leq 2310$; so $13m^2 \leq 4 \cdot 2310^2$, $m \leq 1281$, and with $m = 4c - 1$ this is $c \leq 320$.

4.3 The refutation history (reported, not proved)

The original T4 of the Round 200 dossier read, in full generality:

Proposition 1 (T4 as stated; REFUTED)**.** *An absorbed quad (*$13 \mid h$, $13 \nmid R$*) satisfies* $c \leq 320$.

Round 238's hostile review refuted this as stated. The proof began "$R \mid 30030$, $13 \nmid R \Rightarrow R \mid 2310$," but absorption alone never establishes $R \mid 30030$; that hypothesis comes from non-deadness (R180's Lemma 2, Assumption 1). The reviewer's counterexample:

$$(c, h, a, b) = (321, 13, 1, 16678): \quad a + b = 16679 = 1283 \cdot 13, \tag{4}$$

with $\gcd(ab, h) = 1$ and $R = 16678 \equiv 12 \pmod{13}$, so $13 \nmid R$: this quad is *absorbed* with $c = 321 > 320$ (dead by Lemma 2, but absorbed). In fact $(c, 13, 1, 52c - 14)$ is absorbed for *every* c [5]: the claim "the $13 \mid h$ requirement bounds c by itself" is false. Every other step of the proof survived review ("every remaining step checks out" [5]), so the restored hypothesis in Theorem 4 is exactly the dossier's argument with the missing premise made explicit — and it is the restored version that is machine-checked. The uncorrected Proposition 1 is presented here only as a refuted claim, never as a result.

The same review demoted R200's T2 (trichotomy extension) to **[OPEN]**: the unconditional "$13^1 \| R$ automatically" is false (counterexample $(1, 169, 169, 338)$, for which $R = 2 \cdot 13^2$), and the fiber formula $s \equiv 1 - 4c \pmod{13}$ lacks its stated premise [5]. The conditional core (at a hit, $e - v = 1$ and pinning holds) survived. T3 (fiber-extruded κ-sub-box) **survived** every review step [5] and stays [PROVED*]; its Lean formalization needs the $\kappa/\rho/g_0$ apparatus and is not done here. The 30-quad census (30 absorbed quads with $R \mid 30030$, all dead) stays [COMPUTED] on two parametrizations plus brute force [2].

5 Discussion

What the formalization buys. The three modules turn three previously hand-proved program claims into end-to-end machine-checked theorems with a pinned toolchain and a closed axiom list. The T2 formalization is not a transcription of the hand proof so much as a reduction to finite-set combinatorics: the infinite families of fiber data disappear, leaving "two distinct full-margin subsets of a product $\iff$ two non-singleton margins" — a statement whose mechanization is robust to the framework's evolution. The T5′ kernel likewise isolates the $m = 4c - 1$ arithmetic from the program's κ-box machinery, so the impossibility result no longer depends on any disputed premise.

The review gate works as designed. Three of the four R200 claims put through hostile review failed as stated — all on missing/overbroad hypotheses (unconditional $13^1 \| R$; the unstated $R \mid 30030$) — not on the core mathematics, which survived in every case [5]. The pattern generalizes: Rounds 233/231 had the same failure class (the empty set admitted; nine-element hypotheses omitted) [5]. The correction of T4 shows the gate's intended output: a *smaller true theorem* in place of a larger false one, and the machine-checked version is the corrected one.

Limitations. (i) The T5′ kernel cites the hand-proof machinery as premises (Assumption 1); only the $m = 4c-1$ arithmetic is machine-checked. (ii) R200's T2 remains [OPEN] and T3 awaits the κ-apparatus formalization, so the absorbed regime's structural theorems are not yet fully mechanized. (iii) The R199 suppression census and the $m = 100$ T3 witness are computational, not machine-checked; the "why chain fibers saturate at small moduli" question stays [OPEN] [1]. (iv) The $\bar{\sigma}(m)$ plot (Figure 2) is descriptive: no theorem relates the suppression index to fiber occupancy. (v) The proofs are constructive elementary arithmetic; no new proof method is introduced.

6 Conclusion

We have machine-checked three program results as Lean 4 theorems: the suppression dichotomy and saturation-forcing of Round 199 (`HoleSuppression.t2_iff`, `dichotomy_two_margins`, `t5_saturation` and variants), the absorbed-regime $R \in \{1, 2, 3, 6\}$ impossibility of Round 200 (`AbsorbedRegime.t5prime_combined` and case lemmas), and the corrected self-bound `AbsorbedBound.t4_self_bound` (absorbed + $R \mid 30030 \Rightarrow c \leq 320$). All are [PROVED]: exit-0 builds on two machines, zero sorrys, standard-only axioms. Honestly reported alongside: the refutation of the original T4 (the $(321, 13, 1, 16678)$ counterexample and the all-c absorbed family), the demotion of R200's T2 to [OPEN], and the staged statuses of T3, the 30-quad census, and the R199 surveys. The next mechanization targets are R200's T3 (fiber-extruded slice, review-cleared) and the formalization-pending hand-proved backlog.

7 References

References

[1] D. Blair, *Unsolved Math — Round 199: hole-suppression theory*, program dossier, 2026-10-09. (Supplied datum: T1–T5, survey census, $\bar{\sigma}(m)$ table, computation details.)

[2] D. Blair, *Unsolved Math — Round 200: the* $13 \mid h$ *absorbed regime*, program dossier, 2026-10-09. (Supplied datum: absorbed definition, T2–T5′, 30-quad census, $c \leq 320$ proof.)

[3] D. Blair, *Unsolved Math — Round 230: Lean lane — R222 completion*, program dossier, 2026-10-09. (Supplied datum: R199 T2/T5 and R200 T5′ moved [PROVED]; `bakkie` build exit 0, axioms probes.)

[4] D. Blair, *Unsolved Math — Round 234: Lean lane*, program dossier, 2026-10-09. (Supplied datum: local elaboration exit 0, 8925 jobs.)

[5] D. Blair, *Unsolved Math — Round 238: Lean lane*, program dossier, 2026-10-09. (Supplied datum: hostile review of R200 T2/T3/T4 — T2 fails as stated, T3 survives, T4 fails as stated with counterexample $(321, 13, 1, 16678)$; corrected T4 review-validated; `AbsorbedBound.lean` drafted.)

[6] D. Blair, *Unsolved Math — Round 250: Lane 4 (Lean formalization)*, program dossier, 2026-10-10. (Supplied datum: `AbsorbedBound.lean` repaired and built exit 0 on both machines, zero sorrys, standard-only axioms, md5-identical both sides; corrected T4 [**PROVED**].)

[7] The Lean 4 theorem prover, v4.34.1; Mathlib v4.34.1 (pinned commit `d13f23b723b8a846827a245b89c10fc7d3f11612`).